HOW MUCH FUTURE DO YOU NEED? LOOKAHEAD, CAUSALITY AND PHONE ERROR RATE IN STREAMING ACCENT CONVERSION

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ABSTRACT

Streaming foreign accent conversion (FAC) systems choose a lookahead budget and report a single end-to-end latency, but no published work sweeps that budget or separates the delay a modelling choice imposes from the delay the hardware imposes. We train a causal phone translator at 16 lookaheads from 0 to 640 ms with every other factor held fixed, and report four results. First, quality improves smoothly with lookahead at -0.045 phone error rate (PER) per doubling ($R^2 = 0.98$) with *no locatable knee*: a piecewise fit is preferred under every axis treatment we tried, yet the breakpoint moves between 40 and 180 ms and no bootstrap interval is narrower than three octaves. Instead of a knee we report a *saturation point* at $\approx$240 ms, beyond which each additional 20 ms frame buys less than the measured 2σ noise floor. A 40 ms budget, the value used by the current state of the art, forgoes $\approx 40\%$ of the achievable PER reduction. Second, a pre-registered hypothesis that vowels and approximants degrade faster than obstruents is *not supported*: all seven manner classes improve by 52–72% relative, and the pre-registered grouping explains less variance than the spread within its own groups. It still fails when each error is reweighted by its measured acoustic cost, obtained by forced-aligning per-phone spectral distortion, so the refutation is not an artefact of phone error rate charging every error equally. Third, accent conversion is not merely speech recognition with a different label: on the same dense grid, conversion buys -0.045 PER per doubling of lookahead against -0.029 for accent-faithful transcription at matched capacity, a $1.55\times$ difference, and the gap between the two widens $5.5\times$ from $L{=}0$ to 640 ms. Fourth, per-chunk compute is flat in lookahead (-0.3% to $+1.9\%$ across a $17\times$ change in algorithmic latency, replicated on two architectures), which makes the two latency terms independently actionable. We release the harness, and we document three measurement failures that produced confident but wrong numbers before being caught.

***Index Terms*—** accent conversion, streaming speech, latency, lookahead, self-supervised models

1. INTRODUCTION

Streaming FAC exists and works. PHONOS [1] runs at $\leq$40 ms lookahead and reports an 81% reduction in non-native accent confidence. What does not exist is any account of *why* 40 ms, what it costs in quality, or what any of it does on hardware that is not a datacentre GPU.

Three gaps follow, and this paper addresses the first two directly.

Nobody sweeps the budget at resolution. PHONOS *states* 40 ms and publishes no ablation showing 40 ms is necessary or sufficient; TVTSyn fixes 4 future tokens ($\approx$80 ms) without varying them. DarkStream does ablate, over three points $\{0, 140, 280\}$ ms, and finds the third buys $<$1pp — but three points scored on anonymisation accuracy cannot locate where returns stop, which is the question a deployment budget actually asks. "What would 160 ms buy you" still has no published answer.

Algorithmic and computational latency are reported fused. A figure such as "$<$241 ms on a single GPU" combines a modelling choice with unstated inference and buffering costs. A reader cannot tell whether a faster chip would help. We decompose throughout as

$$t_{\mathrm{e2e}} = \underbrace{c + L}_{t_{\mathrm{alg}}} + t_{\mathrm{cmp}} + t_{\mathrm{buf}}, \tag{1}$$

where c is chunk size, L is lookahead, t_{cmp} is per-chunk compute and t_{buf} is I/O and jitter buffering. t_{alg} is inherent: it is the delay on an infinitely fast machine, because the model refuses to emit frame n until it has seen frame $n{+}L$. t_{cmp} is contingent — quantise, prune, buy a faster chip.

Degradation is reported in aggregate. Existing results are utterance-level MOS, WER or accentedness. Which *sounds* break first is unreported.

Our contributions:

1. A latency–quality characterisation of streaming FAC across 16 lookahead budgets, reporting an exchange rate and a floor-relative saturation point rather than an unidentifiable knee (§5.2).
2. A negative result on a pre-registered phoneme-class hypothesis (§5.3).
3. Evidence that accent conversion needs more lookahead than accent-faithful transcription at matched capacity (§5.4).
4. Two patches that make a masked SSL encoder actually causal, with a truncation proof (§3.2).
5. Three documented measurement failures, each of which produced plausible output before being caught (§6).

2. RELATED WORK

PHONOS [1], TVTSyn [2] and DarkStream [3] are three systems from one group, all pairing FAC or voice conversion with speaker anonymisation. Two of them report more than is sometimes credited, and it is worth being precise about what they do and do not establish. TVTSyn reports CPU as well as GPU latency ($\approx$132 ms and $\approx$79 ms at a 60 ms chunk), so a CPU number for streaming conversion is not unpublished — what is unpublished is a CPU latency–*quality* curve. DarkStream ablates its look-ahead over $\{0, 140, 280\}$ ms and selects 140 ms on the grounds that "an extra 140ms of look-ahead yields a marginal $<$1pp absolute improvement at the expense of doubling the buffering delay". That is a lookahead ablation, on three points, scored on anonymisation accuracy rather than conversion quality.

Table 1. Truncation test on `wavlm-base-plus`. Relative L_2 change in earlier frames after deleting later audio.

configuration	relative L_2	causal?
attention mask only	1.14×10^{-2}	no
+ causal positional conv	6.00×10^{-3}	no
+ cumulative GroupNorm	2.60×10^{-2}	no
both patches	$\mathbf{6.14 \times 10^{-6}}$	**yes**

DarkStream's choice corroborates our saturation point from a different task. Their marginal return collapses somewhere between 140 and 280 ms; we place saturation at ≈240 ms on phone error rate, measured over 16 budgets rather than three. Two systems, two objectives, the same region — which is a stronger position for this paper than the absence of prior evidence would have been. StreamVC [4] reports operation "even on a mobile platform" without naming the device, core count or thread count, which makes the claim unfalsifiable. A recent survey [5] frames the problem space. Related streaming anonymisation work [6, 7, 8] shares the streaming constraint but not the accent objective. Offline FAC [9, 10] and low-latency VC [11] bracket the design space.

Our contribution is orthogonal to all of these: we characterise an axis they parameterise but do not study. We do not claim to beat any of them on quality.

3. METHOD

3.1. A causal phone translator as the probe

Training a full FAC pipeline per lookahead is prohibitive. We instead isolate the component the research question is about. L2-ARCTIC [12] provides, per utterance, both the canonical phone sequence a native speaker would produce (`g2p`) and the sequence the non-native speaker actually produced (`ipa`). Accent conversion is then directly trainable as

non-native audio → native phone sequence,

a CTC head over a frozen causal encoder. This is the supervised core of the task, it costs a GPU-day rather than 200 GPU-hours, and swapping the target tensor from `g2p` to `ipa` yields a matched-capacity control that differs in *one tensor* — a tighter control than AC-versus-VC.

3.2. Making the encoder causal

Masking self-attention in wav2vec2/WavLM-base [13, 14] does *not* produce a causal encoder. Two leaks remain: the positional convolution (`pos_conv_embed`, depthwise, kernel 128, symmetric padding) admits 1.28 s of future at a 20 ms frame rate; and the feature-encoder GroupNorm normalises each channel over the entire utterance, making every output frame depend on every input frame.

We verify causality by truncation: delete audio after frame t and check whether any earlier frame moved (Table 1). Note the third row: patching GroupNorm alone makes matters *worse*. A partial causality fix is not a partial improvement.

3.3. Lookahead is quantised

Lookahead is realised as $\lceil L/f \rceil$ frames at frame rate $f = 20$ ms. A knee narrower than 20 ms is not merely unmeasured but *unrepre-*

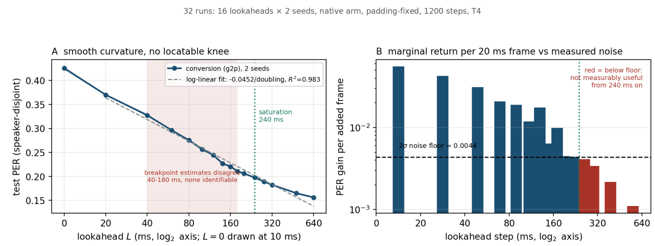

Fig. 1. Dense sweep, conversion arm, 16 lookaheads × 2 seeds. **Left:** smooth curvature; the shaded band spans the breakpoint estimates produced by three defensible treatments of L=0, none identifiable. **Right:** marginal return per added 20 ms frame against the measured 2σ floor; red bars are below it.

sentable, and sampling finer than f silently creates duplicate conditions that reweight any fit. Our dense grid is therefore 20 ms, and we assert zero collisions in t_{alg} across all conditions before fitting.

4. EXPERIMENTAL SETUP

Frozen causal WavLM-base-plus, CTC phone head, 1200 steps, batch 8, chunk 40 ms, 100-frame lookback, speaker-disjoint test split of 400 utterances. Two sweeps: a 7-lookahead × 2-target × 3-seed sweep (42 runs) and a 16-lookahead × 2-seed dense sweep, run on *both* target arms (64 runs), all on an NVIDIA T4. Compute measurements use an Apple M4 and an ARM64 Linux host, CUDA-event or `perf_counter_ns` timed, 28–50 repetitions with three warm-up iterations discarded.

Noise floor. Two sweep processes briefly ran concurrently by operator error, repeating six conditions at *identical seed and configuration*. Those repeats are the only direct measurement of fixed-seed run-to-run noise available, and a multi-seed design cannot produce it: pairing within seed cancels seed effects but not this residual. Differences ranged 1.5×10^{-4} to 6.5×10^{-3} PER, giving $\sigma = 0.0022$ and a 2σ resolvable threshold of $\mathbf{0.0044}$ PER. Every effect below is gated on it.

5. RESULTS

5.1. Compute is flat in lookahead

Across $L \in [0, 640]$ ms at chunk 40 ms on the M4, t_{alg} spans 40–680 ms (17×) while median per-chunk compute moves from 43.23 ms to at most 44.06 ms — a range of -0.3% to $+1.9\%$. On ARM64 the span is 0 to $+7.6\%$. The two terms of Eq. 1 are therefore independently actionable, which is the practical justification for reporting them separately.

5.2. RQ1: an exchange rate and a saturation point, not a knee

Conversion-arm PER falls monotonically from 0.4256 at L=0 to 0.1561 at L=640 ms. Fitting $L \geq 20$ ms on $\log_2 L$ gives

$$\Delta\mathrm{PER} = -0.0452 \text{ per doubling}, \quad R^2 = 0.983, \qquad (2)$$

over 15 points.

There is curvature, and it is not localisable. BIC prefers a two-segment fit under all three axis treatments we tried ($\Delta\mathrm{BIC} = -22.2, -31.2, -36.8$), which by the usual thresholds [15] is strong

Table 2. Conversion-arm PER against lookahead (mean of 2 seeds) and marginal return per added 20 ms frame, in units of the measured 2σ=0.0044 floor. Below 1.0× an additional frame is not measurably useful.

L (ms)	PER	gain/frame	× floor
0	0.4256	—	—
20	0.3701	0.0556	12.7
40	0.3275	0.0426	9.8
60	0.2963	0.0312	7.2
80	0.2754	0.0209	4.8
100	0.2564	0.0190	4.4
120	0.2446	0.0118	2.7
140	0.2269	0.0176	4.0
160	0.2206	0.0064	1.5
180	0.2106	0.0099	2.3
200	0.2062	0.0045	1.0
240	0.1973	0.0044	1.0
280	0.1891	0.0041	**0.9**
320	0.1823	0.0034	**0.8**
480	0.1649	0.0022	**0.5**
640	0.1561	0.0011	**0.3**

evidence. But the breakpoint lands at 40, 180 and 180 ms respectively, and every bootstrap 90% interval spans about three octaves. Sixteen points did not rescue a knee. The reason is mechanical and worth stating: on a $\log_2(L+1)$ axis the $L=0 \rightarrow 20$ ms gap is 4.39 units while every other gap is ≈1.0, so a piecewise fit places a breakpoint just past that gap whatever the data does.

What we report instead is the lookahead beyond which marginal return falls below the measured floor. Per added 20 ms frame, gains run +0.056 (0→20 ms), +0.021 (60→80), +0.0045 (180→200), and are below 0.0044 for every step past 240 ms. We therefore report **saturation at ≈240 ms**, read as 200–280 ms rather than a sharp value: the 140→160 and 180→200 ms steps are already only 1.0–1.5× the floor. Unlike a knee, this quantity depends only on the floor, not on an axis choice.

The cost of a 40 ms budget. PER at L=40 ms is 0.3275; at 240 ms it is 0.1973. A 40 ms budget therefore forgoes **39.7%** of the achievable relative reduction, rising to 52.3% against L=640 ms. "Current budgets are under-provisioned" is thus supportable as diminishing-returns geometry with a measurable saturation point — not as a knee.

5.3. RQ2: manner class is the wrong axis

We pre-registered, before any of this data existed, that vowels and approximants would break first (formant trajectories over 80–250 ms) while stops, fricatives, affricates and nasals would survive short lookahead (locally determined gestures). Testing this on 295,785 reference phones, with each substitution or deletion charged to the class of the *reference* phone:

Group means differ in the predicted direction (0.676 vs 0.636), but that is the weakest available reading and three stronger checks fail. The effect size is 0.66: the between-group difference (0.040) is smaller than the pooled SD across classes (0.060). The surviving group's *internal* spread is 0.190 — 4.7× the between-group difference. And 5 of 12 pairwise orderings are violated, with nasals out-gaining all three breaks-first classes. By L1, only 3 of 6 support even the directional reading and the gain ratio straddles unity (0.92

Table 3. Relative error-rate reduction, L=0 → 640 ms. B = pre-registered as breaking first, S = as surviving. The predicted grouping does not order the data.

	class	rel. gain	ref. phones
S	nasal	0.716	4 020
B	approximant	0.700	4 575
S	fricative	0.687	7 659
B	vowel (mono)	0.673	13 008
B	vowel (diph)	0.654	4 824
S	stop	0.615	7 071
S	affricate	0.525	1 098

Table 4. H2 under acoustic weighting: relative reduction in per-phone distortion, L=0 → 640 ms, 40 utterances, 1451 reference phones per condition. B = pre-registered as breaking first, S = as surviving. The sequence-metric column is computed on the same utterances, so the two differ only in weighting.

	class	acoustic	sequence	phones
B	approximant	0.386	0.825	153
B	vowel (mono)	0.379	0.712	448
S	nasal	0.376	0.784	128
S	fricative	0.372	0.745	272
B	vowel (diph)	0.321	0.667	166
S	stop	0.305	0.663	240
S	affricate	0.257	0.526	44
effect size (need ≥ 1.0)		0.756	0.607	
ordering violations		2/12	4/12	

Chinese to 1.23 Vietnamese).

H2 is not supported. The finding is not that lookahead fails to help vowels — every class improves 52–72% relative, all above the floor. The finding is that manner class does not predict which sounds need right context.

Reweighting each error by what it costs acoustically does not rescue it. PER charges every error exactly 1.0, so a natural objection is that the grouping is fine and PER is simply mis-weighting it: a substituted vowel and a substituted stop cannot sound equally wrong. We tested that directly. Each utterance's canonical phone string is synthesised with a fixed voice and force-aligned to itself with an espeak-IPA CTC model, giving a time span per reference phone; the hypothesis is synthesised, DTW-aligned to the reference, and the spectral distortion inside each phone's own span is charged to that phone's class. The measure is validated before it is used: phones the model got wrong carry 1.72× the distortion of phones it got right (86.8 vs 50.5), so it does detect errors where they occur, and mean distortion falls monotonically with lookahead (71.4 to 46.9, −34%).

The verdict is unchanged (Table 4). Acoustic weighting does move in H2's favour — effect size $0.607 \rightarrow 0.756$ and ordering violations $4/12 \rightarrow 2/12$ against the sequence metric on the same utterances — but not far enough to clear the pre-registered threshold of 1.0, and the failure has the same shape as before: the "survives" group's internal spread (0.120) is 3.5× the difference between the groups (0.034), and a nasal again out-gains a diphthong. So "PER mis-weights the classes" is no longer an available explanation for the sequence-level refutation.

One explanation does remain open, and it is not the one this test closes. Both signals here are TTS renderings of phone strings, so

Table 5. Both arms on the dense grid, mean of 2 seeds. The gap is the transcription arm's PER minus the conversion arm's: near zero at L=0 and widening monotonically with lookahead.

L (ms)	conversion	transcription	gap
0	0.4256	0.4427	+0.0171
20	0.3700	0.3896	+0.0196
40	0.3275	0.3600	+0.0325
80	0.2754	0.3271	+0.0517
160	0.2206	0.2908	+0.0702
240	0.1973	0.2755	+0.0782
320	0.1823	0.2666	+0.0843
640	0.1561	0.2501	+0.0939
exchange rate (PER/doubling)			
conversion	−0.0452	$R^2 = 0.983$	
transcription	−0.0292	$R^2 = 0.984$	

the synthesiser produces canonical formant trajectories for whatever string it is given. This measures the cost of getting a phone's *identity* wrong; it says nothing about a wrong trajectory for a phone whose identity is right. That still needs real converted audio against a native rendition (§7). The sample is also far smaller than the sequence-level test — 1451 phones per condition against 295,785 — and is weighted accordingly.

5.4. RQ3: conversion needs more lookahead than transcription

Identical architecture, capacity, data order, seed and step count; the two arms differ only in the target tensor. Endpoint gain L=0 $\rightarrow$ 640 ms is -0.2695 (63.2% relative) for conversion against -0.1893 (42.8%) for accent-faithful transcription, a paired difference of $+0.0802$ (sd $= 0.0035$, $t = 39.7$).

The dense grid replicates this, and sharpens it. Running the full 16-lookahead × 2-seed sweep on the transcription arm as well puts both arms on the same dense grid (Table 5). The endpoint gains reproduce the 7-point result almost exactly — 63.3% for conversion against 43.5% for transcription, against 63.2% and 42.8% before — on an independent, denser sample. The exchange rates separate cleanly:

$$\text{conversion } -0.0452 \text{ vs transcription } -0.0292 \text{ PER per doubling,}$$

$R^2 = 0.983$ and 0.984 respectively, so **conversion buys $\mathbf{1.55}\times$ as much per doubling of lookahead**.

The mechanism is visible in the gap rather than the levels. At L=0 the two arms are nearly the same task (0.4256 vs 0.4427, a gap of 0.0171); by L=640 ms the gap is 0.0939, **5.5× wider**. It widens in 13 of 15 adjacent steps, and the transcription arm is worse at 16 of 16 lookaheads, every one of them by more than the 0.0044 noise floor. Lookahead, not accuracy, is what separates them — which is the claim H3 makes.

Two secondary points fall out. Both arms saturate at the same ≈240 ms, and neither yields a locatable knee (breakpoints move to $[40, 240, 160]$ ms across axis treatments for transcription, against $[40, 180, 180]$ for conversion), so §5.2's structural conclusion is not an artefact of the conversion arm. And the transcription arm's own curve is just as log-linear, so the shallower slope is a smaller effect of lookahead, not a noisier measurement of the same one.

The claim is carried by the *preference margin* — PER against the other arm's labels minus PER against its own — which grows monotonically in 3/3 seeds for conversion ($+0.032$ at L=0 to $+0.099$ at 640 ms; 18/18 adjacent step × seed comparisons positive) while the transcription control is 0/3 monotone and drifts *down* (-0.0103). Building the claim on a difference rather than a level turned out to matter (§6).

5.5. t_{buf}: jitter measured, I/O not

On the M4, two independent 30 s duplex runs at 40 ms blocks give a required jitter buffer of 0.14 and 0.08 ms, with p99 callback lateness ≈0.06 ms, zero over/underruns and clock drift below $1.3 \times 10^{-4}\%$. The jitter term is **negligible**.

The I/O term is **not measured**, and we report why rather than substituting a number. An acoustic loopback failed on every attempt: the captured level was invariant to output amplitude (peak 0.00505 at amplitude 0.50 versus 0.00468 at 0.95), meaning the speaker contributes no measurable energy to the microphone — consistent with the operating system's echo cancellation, which scales with its reference and therefore cannot be defeated by a louder probe. The driver-reported figure is not a substitute: its excess over the requested block is *identically* 201.69 ms at both 10 and 20 ms blocks and then roughly doubles, i.e. it is queueing, not converter delay.

6. THREE MEASUREMENT FAILURES

Each of the following produced plausible output before being caught. We report them because each is a trap the next person will meet.

A padding bug rewrote the curve's shape. A masked block handed a zero-padded hidden-state tensor to a depthwise convolution and feed-forward that do not respect the key-padding mask. Fixing it was *not* a constant offset: the correction grows with lookahead, from -0.014 PER at L=0 to -0.086 at 320 ms, a spread of 16.4× the noise floor. The buggy curve therefore *understated* the benefit of lookahead, biasing the central quantity of RQ1. A constant offset would have cancelled in every within-run comparison; this did not. The preference margin (§5.4) was unaffected, because the bug moved both PERs together and the margin is a difference.

A breakpoint estimator returns a breakpoint whether or not one exists. ΔBIC cannot distinguish "there is a knee" from "piecewise fits a curve better than a line". Identifiability must be tested by perturbing the x-axis and bootstrapping the location. Our synthetic check reproduces the artefact exactly: on a curve that is log-linear for $L \geq 20$ with L=0 off the extrapolation, including L=0 yields $\Delta\mathrm{BIC} = -39$ and a confident "knee at 40 ms"; dropping it yields +5 and no knee.

An analysis that buckets unmatched symbols into "other" will produce confident output on the wrong alphabet. Our phoneme-class map was keyed on ARPAbet; the sweep emits IPA. Applying one to the other sent 57% of reference tokens, including *every vowel* and /r/, to "other" — and still printed per-class slopes, an H2 verdict and a by-L1 table. Assert coverage; do not infer it from the absence of an error.

7. LIMITATIONS AND REMAINING WORK

Every number here is phone error rate, not perceived quality. No perceptual evaluation was run, so nothing supports a claim about naturalness or audible accentedness: saturation means marginal *PER* gain falls below the noise floor, not that listeners stop noticing improvement.

"Unfunded" would be the wrong reason. Stimuli built from the per-condition outputs cannot carry the judgement: the phone sequences hold no word or stress structure, so even the ground-truth ceiling synthesises as one unsegmented stream, and a system at 0.16–0.42 PER emits no recognisable words to compare. An informal session confirmed both — the listener could not identify what was said, and scored at chance on the anchors. **A funded study on these stimuli would not have measured accentedness at any budget.** It needs converted audio from a synthesis pipeline: future work, not a purchase.

The audio-domain test in §5.3 closes the identity-error half of the phoneme-class question but not the trajectory half, for the same reason. Training is 1200 steps at fixed budget, not to convergence, over a frozen encoder; the dense sweep has no repeated cells, so its noise floor is imported. t_{buf}'s I/O term is unmeasured, an acoustic loopback being defeated by platform echo cancellation, and the embedded class is projected, not measured.

8. CONCLUSION

Lookahead buys accent-conversion quality at a stable exchange rate of ≈ 0.045 PER per doubling, with no knee to locate and measurable returns out to $\approx$240 ms — six times the budget the current state of the art uses, which forgoes about 40% of the achievable reduction. Per-chunk compute is flat across that range, so the trade-off is a modelling decision rather than a hardware one. A pre-registered phoneme-class hypothesis was refuted. We release the harness, the per-condition hypothesis dumps, and the three measurement failures that had to be caught along the way.

9. DATA AND CODE AVAILABILITY

Harness, causality proof, sweep configurations, analysis scripts with their self-tests, per-condition hypothesis dumps for both arms, and host-identity records for every timing are at `https://github.com/Dipeshtripathi13/streaming-fac-lookahead`. Speech data is L2-ARCTIC [12] via the `KoelLabs/L2Arctic` release (3599 annotated utterances, 24 speakers, 6 L1s), CC-BY-NC-4.0. That licence is non-commercial, so training code and configurations are released in place of model weights; they reproduce the reported curves in a few GPU-hours.

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11. COMPLIANCE WITH ETHICAL STANDARDS

This work used only the publicly released L2-ARCTIC corpus and involved no new data collection and no human subjects, so no ethical approval was required. No listening study was run, and no claim about perceived accentedness or naturalness is made. The author declares no competing interests.

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